AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM355
Communications Technology

Final Examination
Spring Semester 2025/2026

Date: 8 /Jun./2026

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Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **50.**
- This exam consists of **6 questions**.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- Calculators are allowed.

Question 1: **(10 marks)**

a) From your study of optical fibre communication systems, you found that there are two mechanisms that can cause pulse spreading: dispersion, and polarisation. Explain briefly how they are caused. (2 marks)

b) A particular antenna in free space receives 18 μW of power at a distance of 5 km from a transmitter. How much will it receive at 15 km if we assume:

i. inverse square law, (2 marks)

ii. inverse fourth-power law? (2 marks)

c) You have studied that one method for transmitting a voice signal is to use a modulation scheme. Suppose you would like to use a half-wave antenna with a length of 1 meter. What carrier frequency would you use? (2 marks)

d) One of the major challenges facing amplitude modulation (AM) systems is their susceptibility to noise. Why is this less of a problem in frequency modulation (FM) and phase modulation (PM) systems? (2 marks)

Question 2: **(8 marks)**

a) For the modulating signal $v_m(t)$ shown in Fig. 1(a) and the carrier signal $v_c(t)$ shown in Fig. 1(b), draw the spectrum of the modulating, carrier, and modulated signals. (3 marks)

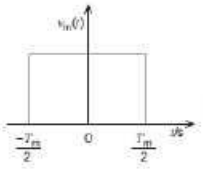


Fig. 1 (a). The modulating signal $v_m(t)$

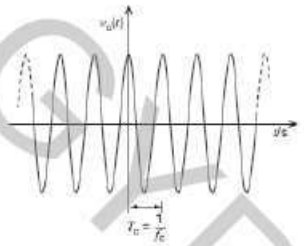


Fig. 1 (b). The carrier signal $v_c(t)$

b) For the rectangular pulse shown in Fig.2 below, draw the corresponding frequency domain representation showing the maximum value of the main lobe and the first pair of spectral nulls. (5 marks)

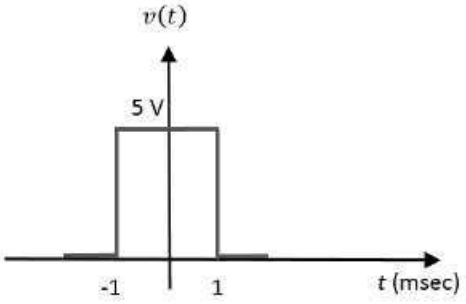


Fig. 2. A rectangular pulse – Question 2

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Question 3: **(8 marks)**

a) Consider a block-interleaving scheme where 30 consecutive (7, 4) hamming codewords are interleaved.

i. The interleaving scheme writes the interleaved codewords into a matrix before sending them (interleaved) through the channel. What are the dimensions of this matrix? (1 mark)

ii. Calculate the latency (in bits) caused by the used block interleaving scheme. (1 mark)

b) A multimedia package contains a simple application for creating music, based on a keyboard with seven notes: A, B, C, D, E, F, G. The developers of the software carry out some testing, and find that the click on the keys towards the centre of the keyboard much more often than the keys at each end. Table below shows the probability of each note being used.

Note	Number of clicks
A	0.03
B	0.19
C	0.20
D	0.21
E	0.17
F	0.16
G	0.04

i. Derive a Huffman code to represent the notes and calculate the average length of the code words. (4 marks)

ii. Calculate the entropy of this source. (2 marks)

Question 4: **(8 marks)**

Starting with an initialised coding table below, follow through the LZW algorithm step by step and complete table below in order to encode the following sequence:

CCACCCAACCCCAA

Note that the initial code for letter **A** is 65, and the corresponding initial code for letter **C** is 67. You should submit full and detailed solution in a table.

Table 1. LZW coding of the sequence

	Input data	P	Q	Code output	New coding table entry	
					Code	Data
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						

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Question 5: **(8 marks)**

a) Discuss the concept of intra-frame compression and explain how it forms the basis of the JPEG image compression standard. (4 marks)

b) Explain the principle of inter-frame compression used in video coding systems such as MPEG. (4 marks)

Question 6: **(8 marks)**

a) With reference to your study of channelisation encoding in WCDMA, assume that the data intended for UE (user equipment) A is: **1 1 0 0**, and the data intended for B is: **0, 0, 0, 0.** The corresponding channelisation code used to encode A's data: **1, −1, 1, −1,** while the corresponding B's code is: **1, −1, −1, 1.** What will be the chip sequence resulting from the simultaneous transmission of A's and B's data? (4 marks)

b) Assume that the frequency bands for ADSL2 over POTS ('plain old telephone system'), that you met in Block 3 (*Principal types of access network: mobile data, DSL broadband, etc.*) of this course, are given as the following:

- ADSL2 upstream band: 25.875 kHz to 138 kHz
- ADSL2 downstream band: 138 kHz to 1.104 MHz

Assuming that the signal-to-noise ratio is constant at 40 dB across the channels, what would the upstream and downstream maximum channel capacities be for the ADSL2? (4 marks)

End of questions

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